

GIRISH KUMAR

Data & AI Engineer | Generative AI Systems & Distributed Data Platforms

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SUMMARY

Data and AI Engineer, 2+ years at Visa, building production Generative AI (GenAI) systems and the distributed infrastructure beneath them: retrieval-augmented generation (RAG) pipelines over ChromaDB and large language model (LLM) APIs, and PySpark/Apache Iceberg lakehouses on an on-premises Hadoop/Kubernetes cluster at 500+ TB historical and 5–10 TB monthly load. ETL/ELT design, orchestration, data modeling, Spark tuning, data quality, end-to-end ownership from design through user acceptance testing (UAT).

TECHNICAL SKILLS

Languages: Python, SQL, Bash/shell scripting, C++, C, JavaScript, HTML, CSS

AI, GenAI: RAG, vector search & embeddings, semantic search, ChromaDB (vector database), LLM APIs (OpenAI, Anthropic, gemini), LangChain, prompt engineering, AI agents, PyTorch, TensorFlow, Transformers, transfer learning, Pandas, NumPy

Data Engineering & Platforms: Apache Spark (PySpark), Apache Airflow, Iceberg, Hadoop/HDFS, Hive Metastore, Trino, Presto, Kubernetes, ETL/ELT, lakehouse architecture, data warehousing, data modeling, data marts, Parquet, partitioning

Databases & Backend: MySQL (pymysql), PostgreSQL, ChromaDB, Flask, REST APIs, React.js, Next.js, Docker

Tools & Practices: Git, GitHub, GitHub Copilot, Cline, Claude Code, DbVisualizer, Postman, Jira, Agile/Scrum, sprint planning

PROFESSIONAL EXPERIENCE

Data Engineer | Visa Inc., Global Data & AI | Bengaluru, India

Jun 2024 – Aug 2026

Project – GenAI Pipeline Automation Platform (RAG, Vector Search, LLM APIs)

- **Cut solution build time from 15 days to 4**, replacing a 15–20-day hand-authoring cycle for multi-module **PySpark** ETL and **Airflow** directed acyclic graphs (DAGs) with a self-serve GenAI web platform (React.js, Flask) featuring automated deployment
- Refactored a monolithic ETL codebase into an installable **Python package** of dependency-ordered sub-packages with class-based module interfaces, **cutting new-solution setup from dozens of script edits to one import** plus a single config file
- Indexed metadata-tagged DAG template chunks in **ChromaDB** so **LLM APIs** assemble runnable, correctly ordered DAGs from semantic retrieval, backed by an LLM config engine that rewrites large JSON configs schema-safely with template fallback
- Automated pre-deployment validation and artifact packaging for generated DAGs, configs, and PySpark modules, and eliminating **6 hours** of manual pre-release checks per cycle through per-step error handling and template-config fallback
- Shipped a multi-turn **RAG chatbot** over DAG source, standard operating procedures (SOPs), and lineage mappings, plus auto-generated STTM and SOP documentation, **so teams could triage without the original pipeline authors**

Project – Conversational Business Intelligence (BI) Lakehouse

- Designed and owned the ETL pipelines behind a natural-language BI agent serving leadership key performance indicators (KPIs) across all global regions, at **500+ TB historical** and **5–10 TB monthly** load on an on-premises Iceberg lakehouse
- Built **PySpark-on-Kubernetes** pipelines over an on-premises **Apache Iceberg** lakehouse (HDFS, Hive Metastore), one parameterised codebase covering backfill and monthly loads, **shipping 20+ KPIs** via a metadata-driven KPI store design
- Root-caused a silent **40–50% card overcount** from a non-additive **COUNT(DISTINCT)** and a **1% year-over-year undercount** from a LEFT JOIN dropping prior-year-only rows, fixing both with window-based flags, salted SUM, and null-safe full outer joins
- **Cut a 7-hour executor-starvation hang to 23 minutes** via dynamic allocation, **collapsed 250+ Spark stages to 25** with **GROUPING SETS** and HDFS checkpointing, and cleared a 759K small-file stall — **overall runtime 7+ hours to under 2 hours**

Project – Quality Control (QC) Automation (Airflow & AI Agents)

- Replaced a 3-day manual QC cycle with **Airflow** DAGs, reconciliation reports, and email alerts, **closing it in under 2 hours** item
- Designed a **multi-agent LLM system** that analyzed **2+ STTM** specs to identify consolidation candidates across ETL base pulls, collapsing multiple pipelines into one and cutting **1 week** of manual spec review migration into **1 day**.

Software Engineering Intern | Visa Inc., Global Solutions Team | Bengaluru, India

May 2023 – Jul 2023

- Built a React.js portal on a Flask REST API with pymysql connectivity, replacing **2–3 days** of manual think-cell deck assembly with real-time generation for a team fielding **100+** requests monthly; adopted into Solution Delivery

EDUCATION

Indian Institute of Technology (IIT) Kharagpur

2019 – 2024

Bachelor of Technology (B.Tech) + Master of Technology (M.Tech) Dual Degree, Computer Science and Engineering

ACADEMIC PROJECTS

Paddy Disease Classification | M.Tech Thesis, TensorFlow

IIT Kharagpur

- Fine-tuned **VGG16, MobileNetV2 & ResNet50V2** on a 20K-image, 15-class corpus from 4 datasets; **88.3% validation accuracy**
- Shipped the winning MobileNetV2 model as a farmer-facing web app: **TensorFlow.js** served via an **Express.js** API and **Next.js** UI

AWARDS & RECOGNITION

- **Significant Contributor Award**, Visa FY25 – accountability, quality and timely delivery on a flagship global data solution
- **Multiple UPLIFT Awards**, Visa FY25 & FY26 – peer-nominated for GenAI expertise, collaboration and knowledge sharing
- **Gold Medal**, Inter-Hall Data Analytics Competition, IIT Kharagpur | **Captain**, Fine Arts & Hockey, IIT Kharagpur – led a gold-winning modeling team and a 14-member Gymkhana hockey squad in General Championship, IIT Kharagpur